

SOC Home Lab

Simulating RDP Brute Force Attack and Detection Using Splunk

Name	Ahmed Saeed Khan
Date	23-June-2026
Tools Used	Splunk Enterprise, Hydra, xfreerdp.
Objective	To simulate a real-world RDP brute force attack and detect it using Splunk SIEM

1. Summary

On 23 June 2026, a simulated RDP brute force attack was carried out in a controlled home lab environment to demonstrate real-world attack techniques and their detection using a SIEM solution. A Kali Linux virtual machine was used as the attacking machine, targeting a Windows 10 virtual machine over Remote Desktop Protocol. Using Hydra, the attacker successfully cracked the target account password and gained unauthorized access via xfreerdp.

Following the initial compromise, the attacker created a backdoor user account, escalated its privileges to Administrator level, and attempted to destroy evidence by clearing the Windows Security event logs.

Throughout the entire attack, Splunk Enterprise — receiving live logs via the Splunk Universal Forwarder — detected and recorded every action in real time. The attacker's attempt to clear local logs proved ineffective, as all evidence had already been forwarded to Splunk. This project highlights the importance of centralized log management and demonstrates how a properly configured SIEM can detect, track, and preserve evidence of an attack even when the attacker attempts to cover their tracks.

2. Tools Used

Tools	Description
Splunk Enterprise	A Security Information and Event Management (SIEM) platform used for collecting, indexing, searching, and analyzing machine-generated data
Splunk Universal Forwarder	A lightweight agent installed on endpoints that collects and forwards log data to a Splunk indexer in real time
Hydra	A fast and flexible open-source password cracking tool that supports numerous protocols including RDP, SMB, and SSH
xfreerdp	An open-source Remote Desktop Protocol client that allows users to connect to Windows machines remotely from Linux

3. Lab Setup

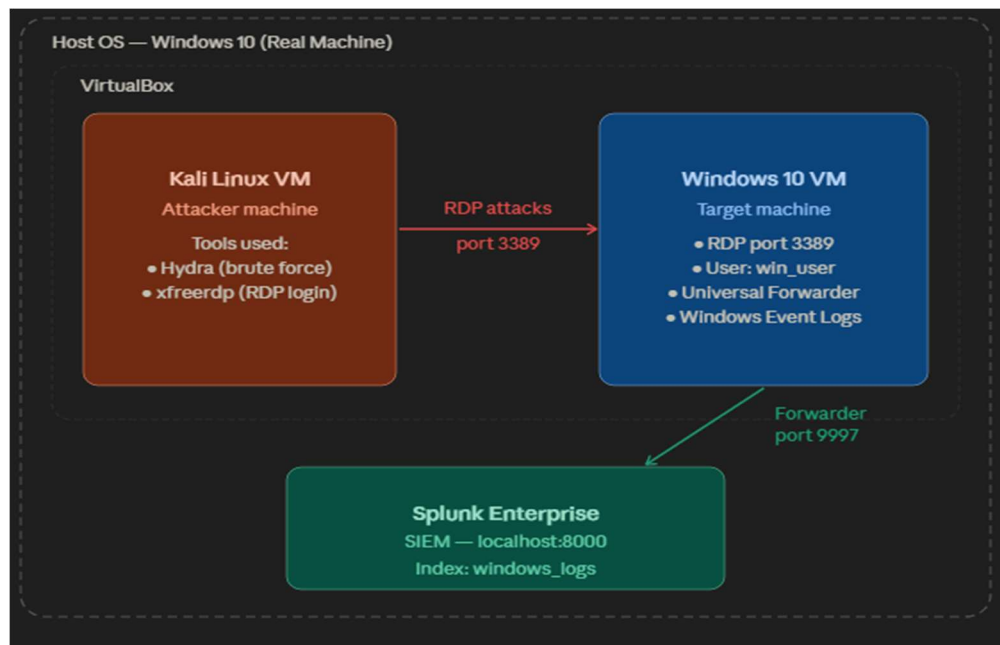
3.1 Environment

The following components were used in this lab:

Field	Value
Host OS	Windows 10 Pro (Real Machine)
SIEM	Splunk Enterprise — localhost:8000
Attacker VM	Kali Linux (VirtualBox)
Target VM	Windows 10 Pro (VirtualBox)
Target VM IP	192.168.100.51
Log Forwarder	Splunk Universal Forwarder
Forwarder Port	9997
Log Index	windows_logs

3.2 Architecture

The lab consists of three components running on a single physical machine. A Kali Linux virtual machine acts as the attacker, while a Windows 10 virtual machine serves as the target. Both VMs run inside VirtualBox on the host Windows 10 machine, where Splunk Enterprise is also installed. The Splunk Universal Forwarder, installed on the Windows 10 VM, continuously ships Windows Security event logs to Splunk in real time, allowing all attacker activity to be monitored and investigated from the host machine.



4. Attack Walkthrough

Phase 1 — RDP Brute Force

The attacker used Hydra, a password cracking tool, to perform a brute force attack against the target machine over RDP (port 3389). The rockyou.txt wordlist was used. Hydra successfully identified the correct password within the first few attempts.

Command used from Kali:

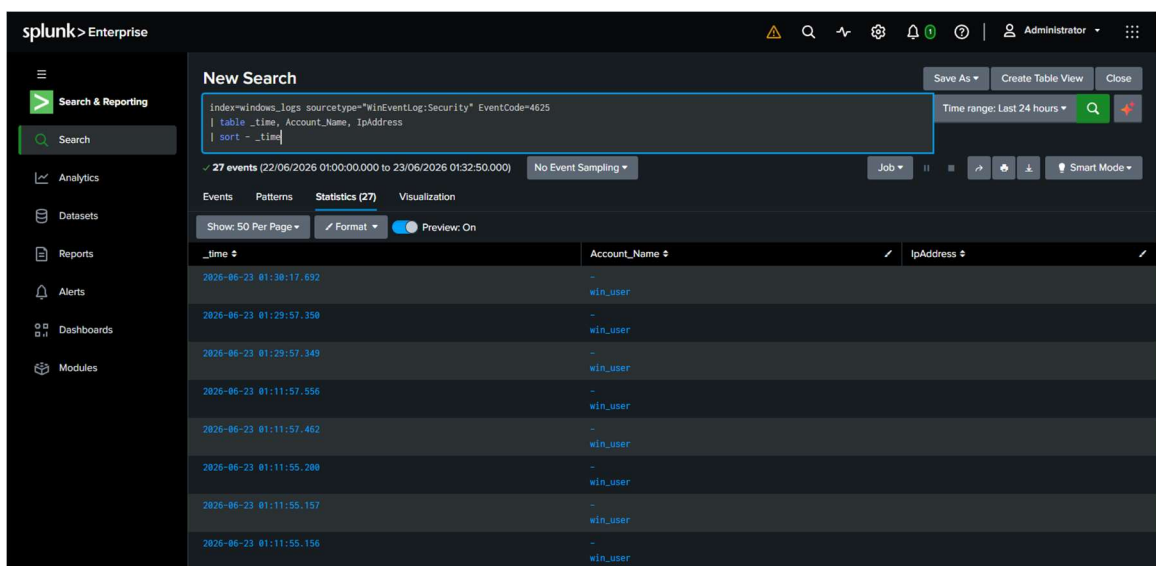
```
hydra -l win_user -P /usr/share/wordlists/rockyou.txt rdp://192.168.100.51 -t 1 -W 5
```

Result:

- Attack started: 2026-06-23 01:11:45
- Attack ended: 2026-06-23 01:30:17
- Total attempts: 27
- Password cracked: 12345
- Target account: win_user
- Attacker machine: kali (Workstation_Name field)

⚠ Note: The IpAddress field appears empty in Event 4625 logs. This is because RDP's Network Level Authentication (NLA) rejects invalid credentials before a full session is established, preventing Windows from logging the source IP.

Splunk detected 27 failed login events (EventCode 4625) against the win_user account:

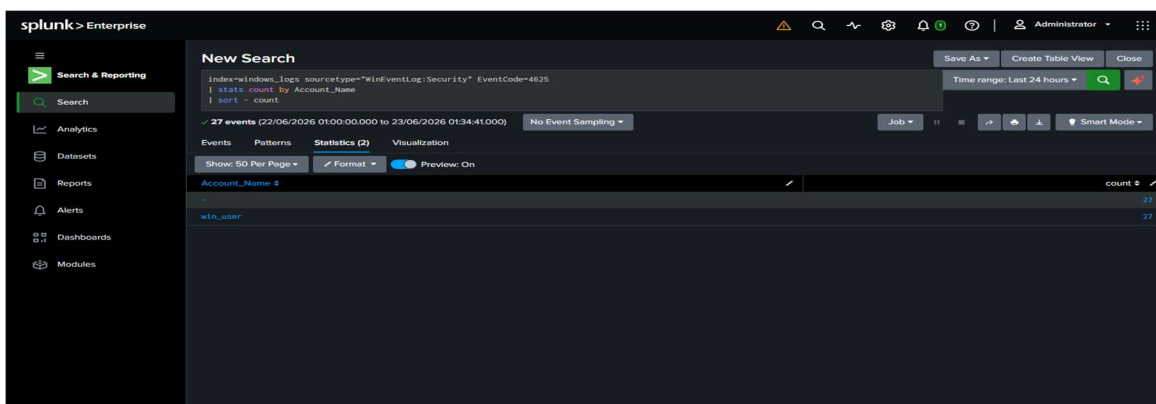


New Search

```
index=windows_logs sourcetype="WinEventLog:Security" EventCode=4625
| table _time, Account_Name, IpAddress
| sort - _time
```

27 events (22/06/2026 01:00:00.000 to 23/06/2026 01:32:50.000) No Event Sampling

_time	Account_Name	IpAddress
2026-06-23 01:38:17.692	win_user	-
2026-06-23 01:29:57.358	win_user	-
2026-06-23 01:29:57.349	win_user	-
2026-06-23 01:11:57.556	win_user	-
2026-06-23 01:11:57.462	win_user	-
2026-06-23 01:11:55.288	win_user	-
2026-06-23 01:11:55.157	win_user	-
2026-06-23 01:11:55.156	win_user	-



New Search

```
index=windows_logs sourcetype="WinEventLog:Security" EventCode=4625
| stats count by Account_Name
| sort - count
```

27 events (22/06/2026 01:00:00.000 to 23/06/2026 01:34:41.000) No Event Sampling

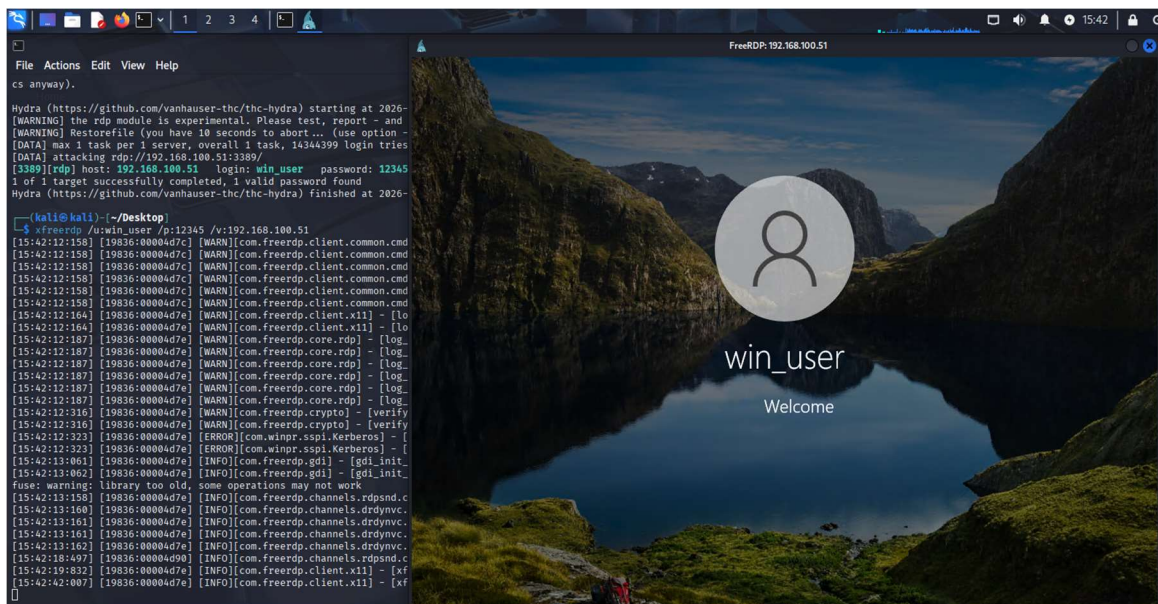
Account_Name	count
win_user	27

Phase 2 — Successful RDP Login

After cracking the password, the attacker used xfreerdp to establish an RDP session to the target machine using the compromised credentials.

Command used from Kali:

```
xfreerdp /u:win_user /p:12345 /v:192.168.100.51
```



Hydra successfully cracking password and xfreerdp establishing RDP session

Splunk detected the successful login as EventCode 4624 with Logon Type 10 (RDP), confirming remote access from the Kali machine:

New Search

```
index=windows_logs sourcetype="WinEventLog:Security" EventCode=4624
| search Account_Name!="*" AND Account_Name!="*" AND Account_Name!="SYSTEM"
| rename Account_Name as User, IpAddress as "Source IP", Logon_Type as "Logon Type"
| table _time, User, "Source IP", "Logon Type"
| sort -_time
```

19 events (22/06/2026 01:00:00.000 to 23/06/2026 01:46:36.000) No Event Sampling

_time	User	Source IP	Logon Type
2026-06-23 01:42:41.832	DWM-3		2
2026-06-23 01:42:41.832	DWM-3		2
2026-06-23 01:42:41.768	LMFD-3		2
2026-06-23 01:42:18.571	win_user		10

Note: The Source IP field is also empty for successful logins. This is a known behavior of the xfreerdp client — it does not populate the IP field in Windows authentication packets.

Phase 3 — Privilege Escalation

Once inside the system, the attacker created a new backdoor user account named 'Attacker' and added it to the local Administrators group to maintain persistent elevated access.

Commands executed on target machine:

```
net user Attacker Password123 /add
net localgroup Administrators Attacker /add
```

```
C:\Windows\system32>net user Attacker Password123 /add
The command completed successfully.

C:\Windows\system32>net localgroup Administrtors Attacker /add
System error 1376 has occurred.

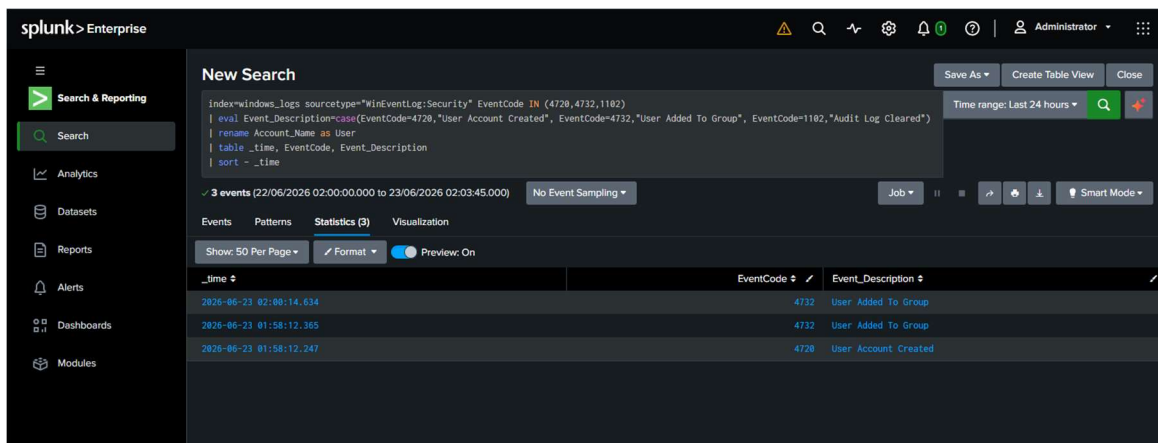
The specified local group does not exist.

C:\Windows\system32>net localgroup Administrators Attacker /add
The command completed successfully.

C:\Windows\system32>_
```

Commands creating backdoor account 'Attacker' and adding to Administrators group

Splunk detected both actions via EventCode 4720 (user created) and EventCode 4732 (user added to group):



Timeline:

- 2026-06-23 01:58:12 — Account 'Attacker' created (EventCode 4720)
- 2026-06-23 01:58:12 — Account added to Administrators group (EventCode 4732)

Phase 4 — Covering Tracks (Log Clearing)

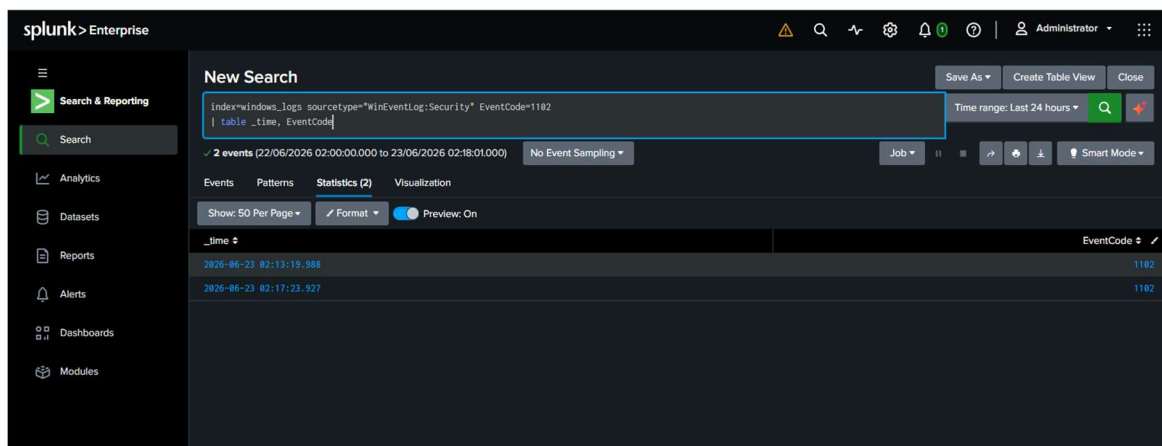
In an attempt to destroy evidence of the attack, the attacker cleared the Windows Security event log using a built-in Windows utility.

Command executed on target machine:

```
wevtutil cl Security
```

```
C:\Windows\system32>wevtutil cl Security
C:\Windows\system32>_
```

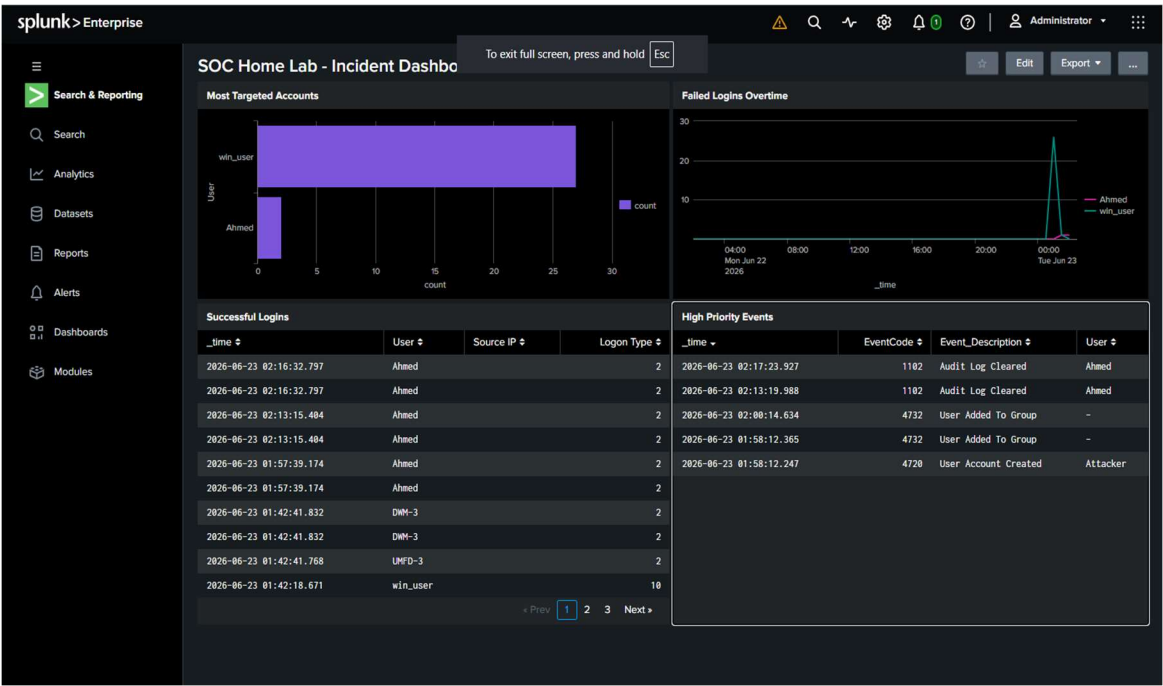
Despite the attacker clearing local logs, Splunk had already received and stored all events. EventCode 1102 (Audit Log Cleared) was captured in Splunk, confirming the clearing attempt itself was logged:



Key Finding: This is exactly why a SIEM exists. The attacker deleted local Windows logs, but Splunk had already shipped and indexed them on the host machine. The attacker's attempt to cover their tracks was itself recorded as evidence.

5. Splunk Dashboard

The SOC Home Lab Incident Dashboard provides a visual summary of all attack phases in a single view:



Splunk Dashboard showing all attack phases: brute force, successful logins, privilege escalation, and log clearing

6. Attack Timeline

Time	Event Code	Description
2026-06-23 01:11:45	4625	Brute force attack started — Hydra begins RDP password attack
2026-06-23 01:11:45 – 01:30:17	4625 x27	27 failed login attempts against win_user from kali
2026-06-23 01:30:17	4625	Last failed attempt recorded — attack phase ends
2026-06-23 01:42:18	4624	Successful RDP login — attacker gains access (Logon Type 10)
2026-06-23 01:58:12	4720	Backdoor account 'Attacker' created

2026-06-23 01:58:12	4732	Account 'Attacker' added to Administrators group
2026-06-23 02:13:19	1102	Windows Security event log cleared by attacker
2026-06-23 02:17:23	1102	Second log clear event detected in Splunk

7. Indicators of Compromise (IOCs)

Field	Value
Attacker Machine	kali (Workstation_Name)
Attack Vector	RDP Brute Force — Port 3389
Tool Used	Hydra + xfreerdp
Target Account	win_user
Password Cracked	12345 (weak password)
Backdoor Account	Attacker
Backdoor Password	Password123
Privilege Level	Local Administrators
EventCodes Triggered	4625, 4624, 4720, 4732, 1102
Total Failed Logins	27

8. Recommendations

Based on the attack observed in this lab, the following security controls are recommended:

- Enforce account lockout policy — lock accounts after 3-5 failed login attempts to prevent brute force attacks
- Disable RDP if not required — if remote access is needed, restrict it to specific trusted IP addresses only
- Enable Multi-Factor Authentication (MFA) — on all remote access methods to prevent credential-based attacks
- Use strong password policies — enforce minimum 12 characters with complexity requirements; ban common passwords

- Monitor EventCode 4625 in real time — alert on 5+ failures within 5 minutes from the same source
- Monitor EventCode 4720 and 4732 — any new account creation or privilege changes should trigger immediate alerts
- Monitor EventCode 1102 — log clearing is always suspicious and should be treated as a critical alert
- Deploy a SIEM — this lab demonstrates that local log deletion is ineffective when a SIEM is in place; logs were forwarded before deletion occurred

9. Conclusion

This lab exercise successfully simulated a complete attack chain — from initial brute force reconnaissance through to privilege escalation and evidence destruction. Every phase of the attack was detected and recorded by Splunk Enterprise, demonstrating the effectiveness of a properly configured SIEM solution.

The most important finding is that the attacker's attempt to clear Windows Security logs was entirely ineffective. The Splunk Universal Forwarder had already shipped all events to the host machine before the logs were deleted. This underscores a fundamental principle of modern security operations: endpoint logs should always be forwarded to a centralized, attacker-inaccessible system in real time.

All detections were performed using Windows Security Event logs and Splunk SPL queries, with no third-party security tools required. This confirms that even a basic SIEM setup provides significant visibility into attacker behavior.